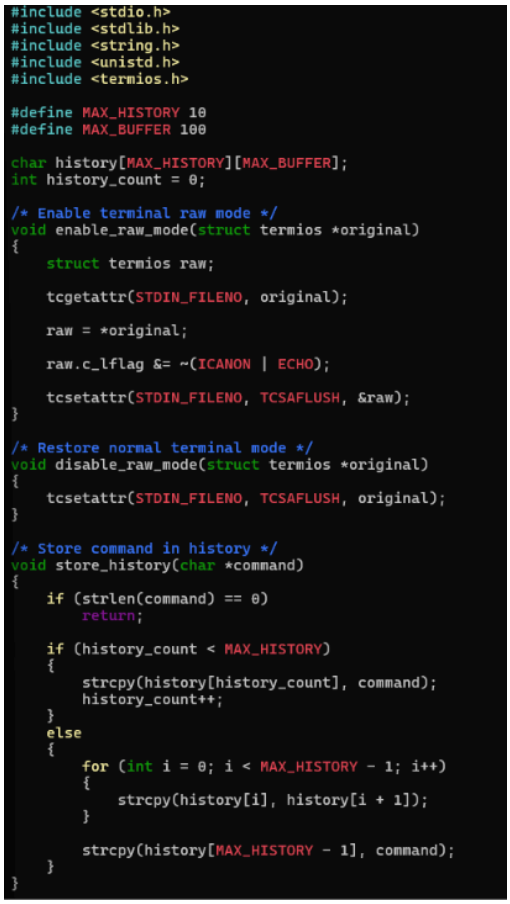
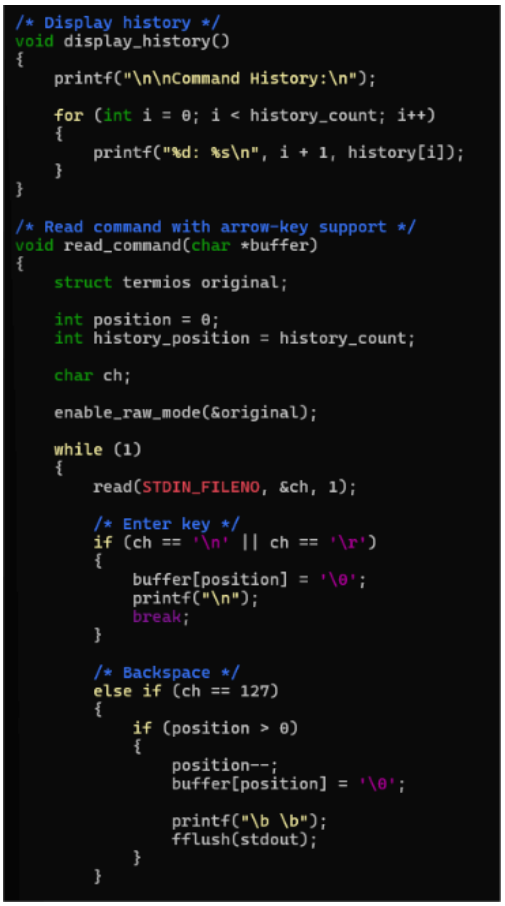
1.To Apply Escape Sequences, Store Command History, Navigate Previous Commands, Navigate Next Commands, Update Input Buffer, Test Recall Functionality.



```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <termios.h>

#define MAX_HISTORY 10
#define MAX_BUFFER 100

char history[MAX_HISTORY][MAX_BUFFER];
int history_count = 0;

/* Enable terminal raw mode */
void enable_raw_mode(struct termios *original)
{
    struct termios raw;

    tcgetattr(STDIN_FILENO, original);

    raw = *original;

    raw.c_lflag &= ~(ICANON | ECHO);

    tcsetattr(STDIN_FILENO, TCSAFLUSH, &raw);
}

/* Restore normal terminal mode */
void disable_raw_mode(struct termios *original)
{
    tcsetattr(STDIN_FILENO, TCSAFLUSH, original);
}

/* Store command in history */
void store_history(char *command)
{
    if (strlen(command) == 0)
        return;

    if (history_count < MAX_HISTORY)
    {
        strcpy(history[history_count], command);
        history_count++;
    }
    else
    {
        for (int i = 0; i < MAX_HISTORY - 1; i++)
        {
            strcpy(history[i], history[i + 1]);
        }

        strcpy(history[MAX_HISTORY - 1], command);
    }
}
```



```c
/* Display history */
void display_history()
{
    printf("\n\nCommand History:\n");

    for (int i = 0; i < history_count; i++)
    {
        printf("%d: %s\n", i + 1, history[i]);
    }
}

/* Read command with arrow-key support */
void read_command(char *buffer)
{
    struct termios original;

    int position = 0;
    int history_position = history_count;

    char ch;

    enable_raw_mode(&original);

    while (1)
    {
        read(STDIN_FILENO, &ch, 1);

        /* Enter key */
        if (ch == '\n' || ch == '\r')
        {
            buffer[position] = '\0';
            printf("\n");
            break;
        }

        /* Backspace */
        else if (ch == 127)
        {
            if (position > 0)
            {
                position--;
                buffer[position] = '\0';

                printf("\b \b");
                fflush(stdout);
            }
        }
```

```
    /* Escape sequence */
    else if (ch == 27)
    {
        char seq[2];

        read(STDIN_FILENO, &seq[0], 1);
        read(STDIN_FILENO, &seq[1], 1);

        /* Up Arrow */
        if (seq[0] == '[' && seq[1] == 'A')
        {
            if (history_position > 0)
            {
                history_position--;

                /* Clear current input */
                while (position > 0)
                {
                    printf("\b \b");
                    position--;
                }

                strcpy(buffer, history[history_position]);

                position = strlen(buffer);

                printf("%s", buffer);
                fflush(stdout);
            }
        }

        /* Down Arrow */
        else if (seq[0] == '[' && seq[1] == 'B')
        {
            if (history_position < history_count - 1)
            {
                history_position++;

                while (position > 0)
                {
                    printf("\b \b");
                    position--;
                }

                strcpy(buffer, history[history_position]);

                position = strlen(buffer);

                printf("%s", buffer);
                fflush(stdout);
            }
```

```
            else
            {
                history_position = history_count;

                while (position > 0)
                {
                    printf("\b \b");
                    position--;
                }

                buffer[0] = '\0';
            }
        }
    }

    /* Normal character */
    else if (ch >= 32 && ch <= 126)
    {
        if (position < MAX_BUFFER - 1)
        {
            buffer[position++] = ch;
            buffer[position] = '\0';

            printf("%c", ch);
            fflush(stdout);
        }
    }
}

disable_raw_mode(&original);
}

int main()
{
    char buffer[MAX_BUFFER];

    printf("=== Command History Program ===\n");
    printf("Type commands and press Enter.\n");
    printf("Use UP/DOWN arrows for history.\n");
    printf("Type 'history' to display history.\n");
    printf("Type 'exit' to quit.\n\n");

    while (1)
    {
        printf("myshell> ");
        fflush(stdout);

        read_command(buffer);

        if (strcmp(buffer, "exit") == 0)
        {
            break;
```

```
        }

        if (strcmp(buffer, "history") == 0)
        {
            display_history();
            continue;
        }

        store_history(buffer);

        printf("Command entered: %s\n", buffer);
    }

    printf("\nProgram terminated.\n");

    return 0;
}
```

```
adithya@LAPTOP-07A78KDI:~/OS$ nano command_history.c
adithya@LAPTOP-07A78KDI:~/OS$ gcc command_history.c -o command_history
adithya@LAPTOP-07A78KDI:~/OS$ ./command_history
=== Command History Program ===
Type commands and press Enter.
Use UP/DOWN arrows for history.
Type 'history' to display history.
Type 'exit' to quit.

myshell> ls
Command entered: ls
myshell> pwd
Command entered: pwd
myshell> date
Command entered: date
myshell> whoami
Command entered: whoami
myshell> myshell> whoami
Command entered: myshell> whoami
myshell> whoami
Command entered: whoami
myshell> date
Command entered: date
myshell> pwd
Command entered: pwd
myshell> ls
Command entered: ls
myshell> history


Command History:
1: ls
2: pwd
3: date
4: whoami
5: myshell> whoami
6: whoami
7: date
8: pwd
9: ls
myshell> ls
```

PART 2 — Dynamic Memory Management

```
adithya@LAPTOP-07A78KDI:~/OS$ nano dynamic_memory.c
adithya@LAPTOP-07A78KDI:~/OS$ gcc dynamic_memory.c -o dynamic_memory
adithya@LAPTOP-07A78KDI:~/OS$ ./dynamic_memory
Dynamic Array Demonstration
Array resized to 10 elements

Array contents:
10 20 30 40 50 60 70 80 90 100

Linked List: 50 -> 40 -> 30 -> 20 -> 10 -> NULL

All dynamically allocated memory released.
adithya@LAPTOP-07A78KDI:~/OS$
```

PART 3 — Verify Using Valgrind

```
adithya@LAPTOP-07A78KDI:~/OS$ valgrind --version
valgrind-3.26.0
adithya@LAPTOP-07A78KDI:~/OS$ gcc dynamic_memory.c -o dynamic_memory
adithya@LAPTOP-07A78KDI:~/OS$ gcc dynamic_memory.c -o dynamic_memory
adithya@LAPTOP-07A78KDI:~/OS$ gcc -g dynamic_memory.c -o dynamic_memory
adithya@LAPTOP-07A78KDI:~/OS$ valgrind --leak-check=full ./dynamic_memory
==1817== Memcheck, a memory error detector
==1817== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==1817== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==1817== Command: ./dynamic_memory
==1817==
Dynamic Array Demonstration
Array resized to 10 elements

Array contents:
10 20 30 40 50 60 70 80 90 100

Linked List: 50 -> 40 -> 30 -> 20 -> 10 -> NULL

All dynamically allocated memory released.
==1817==
==1817== HEAP SUMMARY:
==1817==     in use at exit: 0 bytes in 0 blocks
==1817==   total heap usage: 8 allocs, 8 frees, 1,164 bytes allocated
==1817==
==1817== All heap blocks were freed -- no leaks are possible
==1817==
==1817== For lists of detected and suppressed errors, rerun with: -s
==1817== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
adithya@LAPTOP-07A78KDI:~/OS$
```